Scientific Principles and Applications,

1. Which of the following best describes the Law of Conservation of Energy?
   A) Energy can be created or destroyed but not transferred.
   B) Energy is constantly being used up and replenished.
   C) Energy cannot be created or destroyed, only transformed from one form to another.
   D) Energy is a form of matter that can change states.

2. During cellular respiration, which molecule is primarily broken down to release energy for the cell's activities?
   A) Water
   B) Carbon Dioxide
   C) Glucose
   D) Oxygen

3. When water boils, its temperature remains constant even though heat is continuously being added. What is the primary reason for this phenomenon?
   A) The heat energy is being used to increase the kinetic energy of the water molecules.
   B) The heat energy is being used to overcome the intermolecular forces holding the water molecules together in the liquid phase.
   C) The water molecules are absorbing the heat without changing their energy state.
   D) The boiling point is a fixed temperature that cannot be exceeded.

4. The Greenhouse Effect is a natural process essential for life on Earth. Which of the following best explains its mechanism?
   A) Solar radiation is entirely reflected back into space by the atmosphere.
   B) Certain gases in the atmosphere trap some of the heat radiated from Earth's surface, preventing it from escaping entirely into space.
   C) The ozone layer absorbs all harmful UV radiation, thus warming the planet.
   D) Volcanic eruptions release gases that cool the Earth's surface.

5. Which type of wave requires a medium for its propagation?
   A) Radio waves
   B) Light waves
   C) Sound waves
   D) X-rays

6. In the chemical reaction: 2H , + O , !’ 2H ,O, what does the coefficient '2' in front of H ,O represent?
   A) The atomic number of water.
   B) The number of hydrogen atoms in one water molecule.
   C) The number of water molecules produced.

D) The molar mass of water.

7. The primary function of the circulatory system in the human body is to:
   A) Breakdown food into nutrients.
   B) Exchange gases like oxygen and carbon dioxide.
   C) Transport nutrients, oxygen, hormones, and waste products.
   D) Filter waste products from the blood and produce urine.

8. A scientist wants to test if a new fertilizer increases plant growth. She sets up three groups of plants: Group A receives no fertilizer, Group B receives fertilizer X, and Group C receives the new fertilizer. All groups receive the same amount of water, sunlight, and are of the same plant species. What is Group A in this experiment?
   A) The experimental group
   B) The independent variable
   C) The control group
   D) The dependent variable